

- 2 (a) The grey wolf, *Canis lupus*, is a large predator. During the 20th century, the grey wolf in south-west Europe was hunted almost to extinction.

Fig. 2.1 shows a grey wolf.



Fig. 2.1

- (i) State the genus of the grey wolf.

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- (ii) Suggest **and** explain the effects on the biodiversity of south-west Europe if the grey wolf becomes extinct.

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- (b) Grey wolves can have territories of up to 950 km². Grey wolves can travel up to 1000 km to start a new population.

State reasons why the mark-release-recapture method is **not** suitable for estimating the size of a grey wolf population.

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(c) In 1992, new laws were introduced across south-west Europe to protect the grey wolf.

Fig. 2.2 shows the distribution of grey wolf populations in south-west Europe in 1970 and 2012. No grey wolves from captive-breeding populations were released into the wild in south-west Europe during the period from 1970 to 2012.

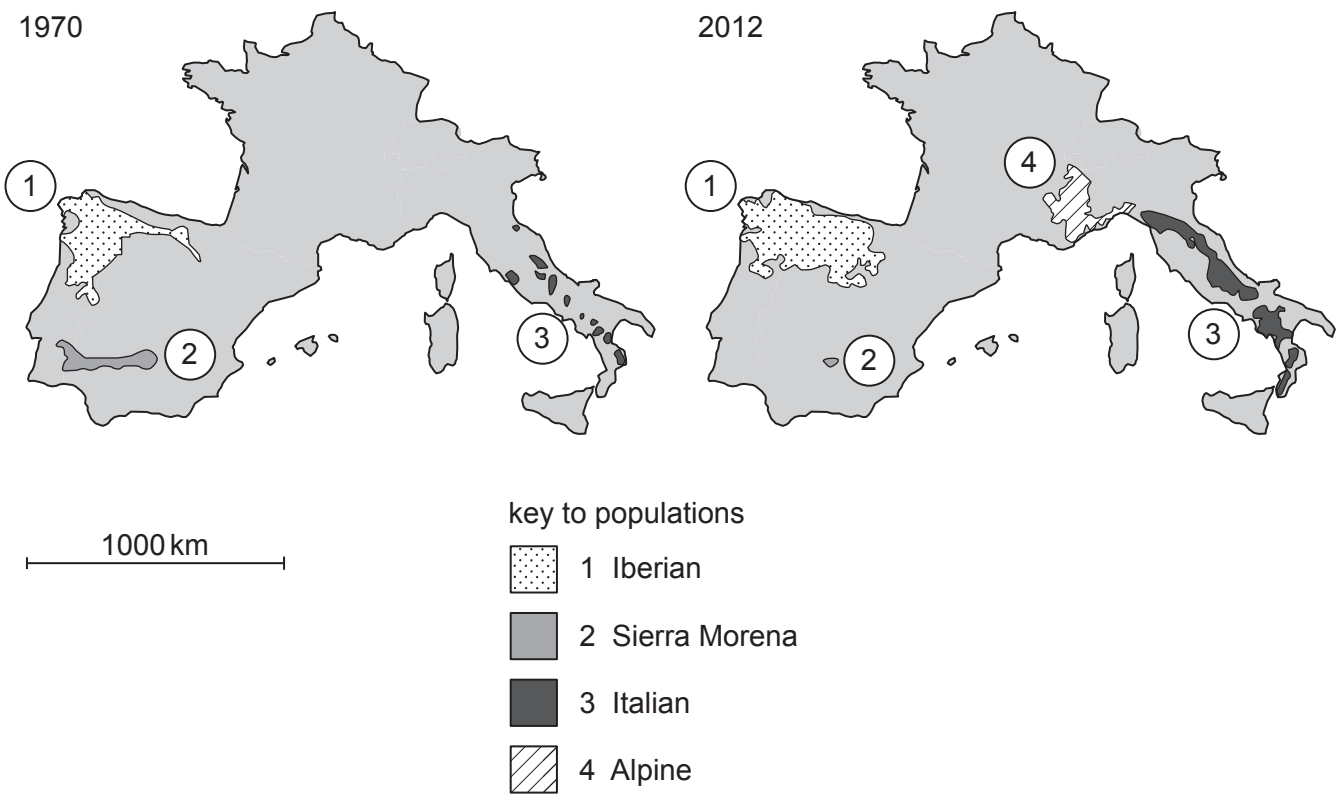


Fig. 2.2

Table 2.1 shows the sizes of the populations of grey wolf shown in Fig. 2.2.

Table 2.1

population	size of population in 1970	size of population in 2012	percentage change
1 Iberian	700	2500	
2 Sierra Morena	60	6	–90
3 Italian	100	800	+700
4 Alpine	0	160	

(i) Complete Table 2.1 to show the percentage change in the size of the Iberian grey wolf population from 1970 to 2012.

Write your answer in the table to the nearest whole number.

- (ii) With reference to Fig. 2.2 and Table 2.1, describe the changes to the grey wolf populations in south-west Europe from 1970 to 2012.

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- (d) (i) In regions of south-west Europe where grey wolf populations are present, farmers are concerned for the safety of their livestock, such as sheep.

Suggest how governments can help farmers who are concerned for the safety of their livestock.

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- (ii) Suggest measures that could help to protect **wild** populations of grey wolves in south-west Europe.

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[Total: 14]